

Examples

1. Mendel's peas.

Mendel's peas had either purple or white flowers. Flower color was due to a single gene with the purple allele (A) dominant to the white allele (a).

The F_1 hybrid (obtained by crossing two pure-breeding lines, one with purple and one with white flowers) has purple flowers and is heterozygous, Aa .

(a) Self the F_1 , pick an F_2 seed at random and grow it up.

i. $\Pr(F_2 \text{ plant has white flowers})$

ii. $\Pr(F_2 \text{ plant has purple flowers})$

iii. $\Pr(F_2 \text{ plant has genotype } AA)$

iv. $\Pr(F_2 \text{ plant has genotype } AA \mid \text{it has purple flowers})$

- (b) Self the F_2 . Grow up ten of the F_3 seeds at random. To save writing, define the following events:

$$\mathbf{P}_2 = \{ F_2 \text{ has purple flowers} \}$$

$$\mathbf{H}_o = \{ F_2 \text{ has genotype } AA \}$$

$$\mathbf{H}_e = \{ F_2 \text{ has genotype } Aa \}$$

$$\mathbf{P}_3 = \{ \text{all ten } F_3 \text{ plants have purple flowers} \}$$

Note that $\mathbf{P}_2 = \mathbf{H}_e$ or $\mathbf{H}_o$. You may want to write out the following, using words. The use of this notation makes things compact but also somewhat harder to follow.

i. $\Pr(\mathbf{P}_2)$

ii. $\Pr(\mathbf{P}_3 \mid \mathbf{H}_o)$

iii. $\Pr(\mathbf{P}_3 \mid \mathbf{H}_e)$

iv. $\Pr(\mathbf{P}_3 \mid \mathbf{P}_2)$

v. $\Pr(\mathbf{H}_o \mid \mathbf{P}_3, \mathbf{P}_2)$

2. The frequency of Down syndrome increases sharply with the age of the mother, and is often a concern among pregnant women age 35 and higher. A common screening test has a false positive rate of about 5% and a false negative rate of about 10%.

(a) The frequency of fetuses with Down syndrome among 33-year-old pregnant women is about $1/420$. Suppose a 33-year-old woman has a positive screening test. What is the chance that she is carrying a child with Down syndrome?

(b) The frequency of fetuses with Down syndrome among 40-year-old pregnant women is about $1/75$. Suppose a 40-year-old woman has a positive screening test. What is the chance that she is carrying a child with Down syndrome?

3. The “Monty Hall” problem.

You’re on a game show, and are presented with three doors. One hides a car; the other two hide goats. You’re allowed to choose a door. Monty Hall then opens one of the other two doors to reveal a goat. You are now allowed to either stick with the door you chose initially, or switch to the other closed door. What should you do (assuming that you are hoping to get the car, and not a goat): stick with your original choice, or switch?

The answer depends on Monty’s behavior.

- (a) If you choose the door with the car, Monty opens one of the other two doors at random. If you choose a door hiding a goat, Monty opens the other door with a goat.**
- (b) Monty Hall opens one of the other two doors at random. If he had revealed the car, you would have lost, but you happen to be in the situation where he revealed a goat.**

4. Mendel, revisited.

Mendel's peas had either purple or white flowers; flower color is due to a single gene, for which the purple allele (A) is dominant to the white allele (a).

We cross two pure-breeding lines (one purple and one white) to produce the F_1 hybrid. We self the F_1 and choose an F_2 seed at random. We grow and self the F_2 and choose two F_3 seeds at random.

Consider the following events.

- $P = \{F_2 \text{ has purple flowers}\}$
- $O = \{F_2 \text{ is homozygous } AA\}$
- $E = \{F_2 \text{ is heterozygous } Aa\}$
- $W = \{F_2 \text{ has white flowers}\}$
- $A_1 = \{F_3 \text{ number 1 has purple flowers}\}$
- $A_2 = \{F_3 \text{ number 2 has purple flowers}\}$

Are A_1 and A_2 independent?